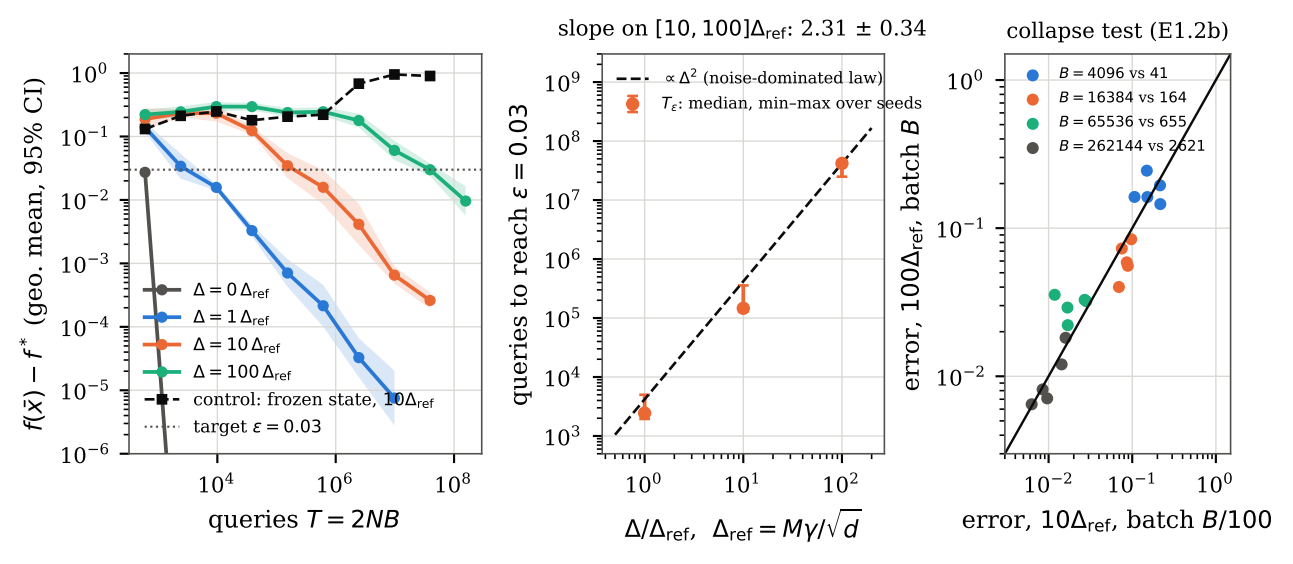
$f(\bar{x}) - f^*$ (geo. mean, 95% CI)
queries $T = 2NB$
$\Delta = 0\,\Delta_{\mathrm{ref}}$
$\Delta = 1\,\Delta_{\mathrm{ref}}$
$\Delta = 10\,\Delta_{\mathrm{ref}}$
$\Delta = 100\,\Delta_{\mathrm{ref}}$
control: frozen state, $10\Delta_{\mathrm{ref}}$
target $\varepsilon = 0.03$
slope on $[10, 100]\Delta_{\mathrm{ref}}$: 2.31 ± 0.34
$\propto \Delta^2$ (noise-dominated law)
T_ε: median, min–max over seeds
queries to reach $\varepsilon = 0.03$
$\Delta/\Delta_{\mathrm{ref}}$, $\Delta_{\mathrm{ref}} = M\gamma/\sqrt{d}$
collapse test (E1.2b)
$B = 4096$ vs 41
$B = 16384$ vs 164
$B = 65536$ vs 655
$B = 262144$ vs 2621
error, $100\Delta_{\mathrm{ref}}$, batch B
error, $10\Delta_{\mathrm{ref}}$, batch $B/100$